MILITARY SPECIFICATION

TIRE SCRIBER FOR AUTOMOTIVE WHEEL ALIGNMENT

This specification is approved for use within the US Army Armament Research, Development and Engineering Center, Department of the Army and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of automotive, heavy duty truck wheel alignment tire scriber to be used for inscribing a guide line on the tread of tires for wheel alignment on vehicles having wheel sizes up to 12:00 by 24 inches, and a tread width range of 48 inches through 84 inches.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock - Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

MILITARY

MIL-P-116 - Preservation-Packaging, Method of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernmental documents which is current on the date of the solicitation.

Uniform classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Freight Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

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3.2 Design. The heavy duty truck tire scriber shall be new and of the manufacturer's current commercial model and design. The scriber design shall incorporate all the requirements specified herein and to the best commercial practices of the wheel alignment equipment manufacturing industry.

3.2.1 Maintainability. All parts subject to wear, breakage, or distortion shall be easily accessible for adjustment, repair, or replacement.

3.3 Construction. All parts and materials used in fabrication of the tire scriber shall be new. In no event shall patching processes be used for reclaiming defective parts. All surfaces of all parts shall be free of burrs, defects, and dirt. Parts subject to removal or adjustment shall not be permanently deformed by the assembly process or normal use.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practices within the wheel alignment equipment manufacturing industry, shall be suitable for the intended purpose in the design of the end item, and shall be free from defects which would adversely affect the performance accuracy, durable usable life, and maintainability of the individual components. The overall suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889. The heavy duty truck tire scriber may contain reclaimed materials provided the materials will not jeopardize the intended use or performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the heavy duty truck scriber.

3.3.2 Workmanship. Workmanship of the scriber, accessories and ancillary items shall be of a quality equal to that of the best commercial practices of the wheel alignment equipment manufacturing industry.

3.3.3 Threads. All machines threads shall conform to FED-STD-H28 and the applicable detailed standards referenced therein.

3.3.4 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed or soldered parts. These operations shall not be employed as repair measure for defective parts.

3.3.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for adjustment and preventing loss of tightness. All parts subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.6 Surfaces. Surfaces of all castings, forgings, molded parts, stampings and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, flash and other harmful or extraneous materials.

3.3.7 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances. All parts of the end item shall be interchangeable with their respective repair/replacement parts without requiring modification of the part or its mating components. All parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part

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number and shall be identified in the repair parts manual. All replacement, repair and spare parts shall provide the same or better form, fit, function, and performance as the parts provided initially installed in the end item.

3.4 Components. The tire scriber shall consist of the following principal component parts.

3.4.1 Tire scriber. The tire scriber shall consist of a stand equipped with a spring loaded moveable arm. The arm shall be fitted with a hardened steel replaceable pointer or wheel for scribing a hairline mark in a true plane of wheel rotation, on the solid portion of the tire tread surface as the wheel is rotated on its axle/spindle . CAUTION: Replaceable point may drop out when slotted machine screw is removed (nonfriction fit).

3.5 Painting/plating. All components with the exception of the scriber and stands shall be plated for protection against corrosion. The painting and material preparation shall be in accordance with the best commercial practices of the wheel alignment equipment manufacturing industry. Immediately prior to painting, all surfaces shall be cleaned and dried and free of all foreign matter. The protecting paint coating shall be not less than 2.0 mil in thickness. The paint adhesion shall be such that no paint shall be peeled by fiberglass tape when two intersecting cuts expose the base metal for a distance of not less than 2 inches. The surface hardness of the paint shall be between the industrial standards of H to HB.

3.6 Identification of product. The tire scriber shall be marked for identification in accordance with MIL-STD-130 and shall include the National Stock Number.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When specified, first article inspection shall be performed on the preproduction models) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.5, and all tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of a through c as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.5).
b. Protective finish (see 4.6.1).
c. Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following:

Inspection or test	Level	AQL
Product examination (see 4.5)	II	1.0
Protective finish (see 4.6.1	S-4	1.0
Packaging inspection (see 4.7)	S-4	4.0

4.5 Product examination. Visually, dimensionally, and manually examine each tire scriber to determine conformance with the requirements of 3.2 through 3.4.1 and 3.6. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examination includes measuring dimensions including weights, as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the tire scriber and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.3.7), shall be cause for rejection.

4.6 Tests.

4.6.1 Paint test. The hardness, adhesion and thickness of the paint used on the tester shall be checked as follows (see 3.5):

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the tester until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the tester, shall have a sharpened H lead pencil pressed into the paint as in a. Failure of the surface to pit (paint too hard) shall be rejected.

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c. Two two-inch intersecting lines shall be cut through the paint to the metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection.

4.7 Packaging inspection. The tester shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5.

5 PACKAGING

5.1 Preservation. The preservation shall be level A or Level C, as specified in the contract (see 6.2.1).

5.1.1 Level A. Scribers shall be disassembled the minimum necessary to accomplish reduction in cube. Unpainted or unplated ferrous metal surfaces, including threaded surfaces and surfaces exposed by disassembly shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116. Each scriber complete with technical data shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. Technical data shall be packaged method IC-1 of MIL-P-116 and placed with the item in the appropriate container.

5.1.2 Level C. The scribers shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. The packing shall be level A, B, or C, as specified in the contract (see 6.2.1).

5.2.1 Level A. The unit packaged scribers shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type; or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes or damaging of the contents in accordance with MIL-STD-1186.

5.2.2 Level B. The unit packaged scribers shall be packed in close-fitting boxes conforming to PPP-B-591, class I; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-640, class 1, grade B. The contents shall be cushioned, blocked and braced to prevent movement within the boxes or damaging of the contents.

5.2.3 Level C. The unit packaged scribers shall be packed in accordance with MIL-STD-1190 in a manner which will ensure arrival at destination in satisfactory condition and which will be acceptable to the carrier.

5.3 Marking. All marking for levels A, B, and C shall be in accordance with MIL-STD-129. Bar coding is required and shall be in accordance with MIL-STD-129 (see 6.2.1).

6 NOTES

6.1 Intended use. This specification covers tire scriber intended for inscribing a guideline on the tread of tires for wheel alignment.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article, when required (see 3.1).
c. Selection of applicable level of preservation, packing and packing required (see 5.1, 5.2, and 5.3).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instruction for operations and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 Subject term (key word) listing.

AMSC number
Design
First Article
Tire scriber

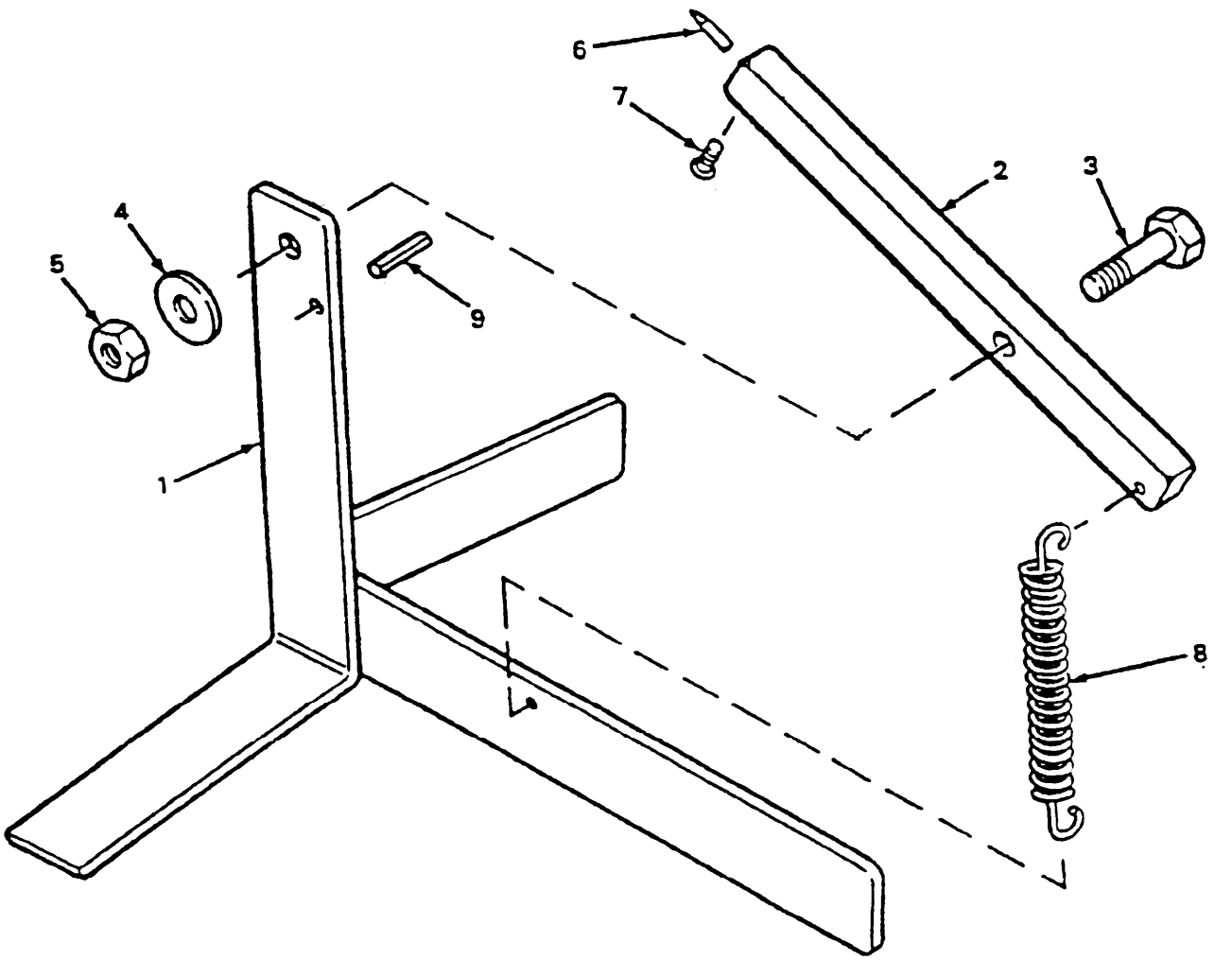
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Army - AL

Review activities:

Preparing activity:
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Project No. 4910-A680

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NOTES:

Item No.	Description	Qty.
1.	Stand	1
2.	Arm	1
3.	1/4"-20 x 3/4" Hex Head Cap Screw	1
4.	1/4" Flat Washer	1
5.	1/4"-20 Lock Nut	1
6.	Pointer Pin	1
7.	#10-24 x .38" Slotted Head Machine Screw	1
8.	Spring	1
9.	1/8" Dia. x 1/2" Roll Pin	1

FIGURE 1. TIRE SCRIBER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-64053 Tire Scriber for Automotive Wheel Alignment

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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